

2Министерство науки и высшего образования Российской Федерации
Муромский институт (филиал)
федерального государственного бюджетного образовательного учреждения высшего образования
«Владимирский государственный университет
Имени Александра Григорьевича и Николая Григорьевича Столетовых»
(МИВлГУ)

Факультет _____ ИТР
Кафедра _____ ПИИ

ЛАБОРАТОРНАЯ РАБОТА №10

по _____ Разработка кроссплатформенных приложений
Тема _____ Интернационализация приложения

Руководитель

Кульков Я.Ю.

(фамилия, инициалы)

(подпись)

(дата)

Студент ПИИ-124

(группа)

Кочедыков Е.А.

(фамилия, инициалы)

(подпись)

(дата)

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Лабораторная работа №10

Тема: Интернационализация приложения

Цель работы: изучить способы интернационализации приложения

Задание.

1. Разработать для своего приложения три файла ресурсов: английский, русский и третий язык по выбору. Способ выбора языка приложения либо через файл ресурсов, либо в самом приложении.
2. Вынести все SQL-выражения в отдельные файлы.
3. Все основные действия в приложении логировать.
4. Реализовать аутентификацию пользователя и подключение к БД.

Ход работы:

Листинг 1. GiveMaterialController.java

```
package ru.kochedykovdev.laba7.controllers;

import javafx.beans.binding.Bindings;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.scene.Node;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.DatePicker;
import javafx.scene.control.ListCell;
import javafx.scene.control.TextField;
import javafx.stage.Stage;
import ru.kochedykovdev.laba7.dao.BuildingObjectDao;
import ru.kochedykovdev.laba7.dao.InvoiceDao;
import ru.kochedykovdev.laba7.dao.InvoiceItemDao;
import ru.kochedykovdev.laba7.dao.MaterialCardDao;
import ru.kochedykovdev.laba7.dao.MaterialStockDao;
import ru.kochedykovdev.laba7.model.BuildingObject;
import ru.kochedykovdev.laba7.model.Invoice;
import ru.kochedykovdev.laba7.model.InvoiceItem;
import ru.kochedykovdev.laba7.model.MaterialCard;
import ru.kochedykovdev.laba7.model.MaterialStock;
import ru.kochedykovdev.laba7.util.Messages;
import ru.kochedykovdev.laba7.util.FieldValidation;
```

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Провер.		Кульков Я.Ю.								МИВУ ПИН - 124				
Н.контр.														
Утв.														

```

import java.math.BigDecimal;
import java.sql.SQLException;
import java.time.LocalDateTime;
import java.util.Optional;

public class GiveMaterialController {

    private final InvoiceDao invoiceDao;
    private final InvoiceItemDao invoiceItemDao;
    private final BuildingObjectDao buildingObjectDao;
    private final MaterialCardDao materialCardDao;
    private final MaterialStockDao materialStockDao;

    @FXML
    private TextField invoiceNumberField;
    @FXML
    private ComboBox<BuildingObject> objectComboBox;
    @FXML
    private DatePicker issueDatePicker;
    @FXML
    private ComboBox<MaterialCard> materialComboBox;
    @FXML
    private TextField quantityField;
    @FXML
    private TextField priceField;
    @FXML
    private Button submitButton;

    public GiveMaterialController(InvoiceDao invoiceDao,
                                InvoiceItemDao invoiceItemDao,
                                BuildingObjectDao buildingObjectDao,
                                MaterialCardDao materialCardDao,
                                MaterialStockDao materialStockDao) {

        this.invoiceDao = invoiceDao;
        this.invoiceItemDao = invoiceItemDao;
        this.buildingObjectDao = buildingObjectDao;
        this.materialCardDao = materialCardDao;
        this.materialStockDao = materialStockDao;
    }

    @FXML
    private void initialize() {
        try {
            setupCombo(objectComboBox);

```

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```

        setupCombo(materialComboBox);
        objectComboBox.getItems().setAll(buildingObjectDao.findAll());
        materialComboBox.getItems().setAll(materialCardDao.findAll());
        issueDatePicker.setValue(java.time.LocalDate.now());

        FieldValidation.onlyDecimal(quantityField);
        FieldValidation.onlyDecimal(priceField);

        FieldValidation.bindSubmit(submitButton, Bindings.createBooleanBinding(
            () -> !invoiceNumberField.getText().isBlank()
                && objectComboBox.getValue() != null
                && materialComboBox.getValue() != null
                && issueDatePicker.getValue() != null
                && isPositive(quantityField.getText())
                && isPositive(priceField.getText()),
            invoiceNumberField.textProperty(),
            objectComboBox.valueProperty(),
            materialComboBox.valueProperty(),
            issueDatePicker.valueProperty(),
            quantityField.textProperty(),
            priceField.textProperty()
        ));
    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

@FXML
private void onCloseClick(ActionEvent event) {
    closeWindow(event);
}

@FXML
private void onSubmitClick(ActionEvent event) {
    try {
        MaterialCard material = materialComboBox.getValue();
        BigDecimal qty = new BigDecimal(quantityField.getText().trim());

        MaterialStock stock = findStock(material.getId())
            .orElseThrow(() -> new IllegalStateException("Материала нет на
складе"));

        if (stock.getQuantity().compareTo(qty) < 0) {
            throw new IllegalStateException("Недостаточно материала на
складе");
        }
    }
}

```

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```

    }

    Invoice invoice = new Invoice(
        null,
        invoiceNumberField.getText().trim(),
        objectComboBox.getValue().getId(),
        issueDatePicker.getValue()
    );

    invoiceDao.insert(invoice);

    InvoiceItem item = new InvoiceItem(
        null,
        invoice.getId(),
        material.getId(),
        qty,
        new BigDecimal(priceField.getText().trim())
    );

    invoiceItemDao.insert(item);

    stock.setQuantity(stock.getQuantity().subtract(qty));
    stock.setLastUpdated(LocalDateTime.now());
    materialStockDao.update(stock);

    closeWindow(event);
} catch (Exception e) {
    showError(e.getMessage());
}
}

private Optional<MaterialStock> findStock(Long materialId) throws SQLException
{
    return materialStockDao.findAll().stream()
        .filter(s -> materialId.equals(s.getMaterialId()))
        .findFirst();
}

private static boolean isPositive(String text) {
    if (text.isBlank()) {
        return false;
    }
    return new BigDecimal(text.trim()).compareTo(BigDecimal.ZERO) > 0;
}

private static <T> void setupCombo(ComboBox<T> comboBox) {
    comboBox.setCellFactory(list -> new ListCell<>() {

```

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```

        @Override
        protected void updateItem(T item, boolean empty) {
            super.updateItem(item, empty);
            setText(empty || item == null ? null : item.toString());
        }
    });
    comboBox.setButtonCell(comboBox.getCellFactory().call(null));
}

private void closeWindow(ActionEvent event) {
    Stage stage = (Stage) ((Node) event.getSource()).getScene().getWindow();
    stage.close();
}

private void showError(String message) {
    Alert alert = new Alert(Alert.AlertType.ERROR);
    alert.setHeaderText(Messages.bundle.getString("error.header"));
    alert.setContentText(message);
    alert.showAndWait();
}
}

```

Листинг 2. GiveToolController.java

```

package ru.kochedykovdev.laba7.controllers;

import javafx.beans.binding.Bindings;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.scene.Node;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.DatePicker;
import javafx.scene.control.ListCell;
import javafx.scene.control.TextField;
import javafx.stage.Stage;
import ru.kochedykovdev.laba7.dao.BuildingObjectDao;
import ru.kochedykovdev.laba7.dao.ToolDao;
import ru.kochedykovdev.laba7.dao.ToolIssueDao;
import ru.kochedykovdev.laba7.model.BuildingObject;
import ru.kochedykovdev.laba7.model.Tool;
import ru.kochedykovdev.laba7.model.ToolIssue;
import ru.kochedykovdev.laba7.util.Messages;
import ru.kochedykovdev.laba7.util.FieldValidation;

```

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```

import java.sql.SQLException;

public class GiveToolController {

    private final ToolDao toolDao;
    private final BuildingObjectDao buildingObjectDao;
    private final ToolIssueDao toolIssueDao;

    @FXML
    private ComboBox<Tool> toolComboBox;
    @FXML
    private ComboBox<BuildingObject> objectComboBox;
    @FXML
    private TextField issuedToField;
    @FXML
    private DatePicker issueDatePicker;
    @FXML
    private DatePicker returnDatePicker;
    @FXML
    private Button submitButton;

    public GiveToolController(ToolDao toolDao,
                              BuildingObjectDao buildingObjectDao,
                              ToolIssueDao toolIssueDao) {

        this.toolDao = toolDao;
        this.buildingObjectDao = buildingObjectDao;
        this.toolIssueDao = toolIssueDao;
    }

    @FXML
    private void initialize() {
        try {
            setupCombo(toolComboBox);
            setupCombo(objectComboBox);
            toolComboBox.getItems().setAll(toolDao.findAll());
            objectComboBox.getItems().setAll(buildingObjectDao.findAll());
            issueDatePicker.setValue(java.time.LocalDate.now());

            FieldValidation.bindSubmit(submitButton, Bindings.createBooleanBinding(
                () -> toolComboBox.getValue() != null
                    && objectComboBox.getValue() != null
                    && !issuedToField.getText().isBlank()
                    && issueDatePicker.getValue() != null
                    && returnDatePicker.getValue() != null
            ));
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

```

```

        &&
!returnDatePicker.getValue().isBefore(issueDatePicker.getValue()),
        toolComboBox.valueProperty(),
        objectComboBox.valueProperty(),
        issuedToField.textProperty(),
        issueDatePicker.valueProperty(),
        returnDatePicker.valueProperty()
    ));
} catch (SQLException e) {
    showError(e.getMessage());
}
}

@FXML
private void onCloseClick(ActionEvent event) {
    closeWindow(event);
}

@FXML
private void onSubmitClick(ActionEvent event) {
    try {
        ToolIssue issue = new ToolIssue(
            null,
            toolComboBox.getValue().getId(),
            objectComboBox.getValue().getId(),
            issuedToField.getText().trim(),
            issueDatePicker.getValue(),
            returnDatePicker.getValue(),
            null
        );
        toolIssueDao.insert(issue);
        closeWindow(event);
    } catch (Exception e) {
        showError(e.getMessage());
    }
}

private static <T> void setupCombo(ComboBox<T> comboBox) {
    comboBox.setCellFactory(list -> new ListCell<>() {
        @Override
        protected void updateItem(T item, boolean empty) {
            super.updateItem(item, empty);
            setText(empty || item == null ? null : item.toString());
        }
    });
}

```

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```

        comboBox.setButtonCell (comboBox.getCellFactory().call (null));
    }

    private void closeWindow(ActionEvent event) {
        Stage stage = (Stage) ((Node) event.getSource()).getScene().getWindow();
        stage.close();
    }

    private void showError(String message) {
        Alert alert = new Alert(Alert.AlertType.ERROR);
        alert.setHeaderText(Messages.bundle.getString("error.header"));
        alert.setContentText(message);
        alert.showAndWait();
    }
}

```

Листинг 3. MainController.java

```

package ru.kochedykovdev.laba7.controllers;

import javafx.beans.property.SimpleStringProperty;
import javafx.collections.FXCollections;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.fxml.FXMLLoader;
import javafx.scene.Node;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.scene.control.TabPane;
import javafx.scene.control.TableColumn;
import javafx.scene.control.TableView;
import javafx.scene.control.TextField;
import javafx.stage.Modality;
import javafx.stage.Stage;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import ru.kochedykovdev.laba7.dao.BuildingObjectDao;
import ru.kochedykovdev.laba7.dao.InvoiceDao;
import ru.kochedykovdev.laba7.dao.InvoiceItemDao;
import ru.kochedykovdev.laba7.dao.MaterialCardDao;
import ru.kochedykovdev.laba7.dao.MaterialReceiptDao;
import ru.kochedykovdev.laba7.dao.MaterialReturnDao;
import ru.kochedykovdev.laba7.dao.MaterialStockDao;
import ru.kochedykovdev.laba7.dao.ProrabDao;
import ru.kochedykovdev.laba7.dao.SupplierDao;

```

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```

import ru.kochedykovdev.laba7.dao.ToolDao;
import ru.kochedykovdev.laba7.dao.ToolIssueDao;
import ru.kochedykovdev.laba7.model.BuildingObject;
import ru.kochedykovdev.laba7.model.Invoice;
import ru.kochedykovdev.laba7.model.InvoiceItem;
import ru.kochedykovdev.laba7.model.MaterialCard;
import ru.kochedykovdev.laba7.model.MaterialReceipt;
import ru.kochedykovdev.laba7.model.MaterialReturn;
import ru.kochedykovdev.laba7.model.MaterialStock;
import ru.kochedykovdev.laba7.model.Prorab;
import ru.kochedykovdev.laba7.model.Supplier;
import ru.kochedykovdev.laba7.model.Tool;
import ru.kochedykovdev.laba7.model.ToolIssue;
import ru.kochedykovdev.laba7.util.Messages;

import java.io.IOException;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.HashMap;
import java.util.LinkedHashMap;
import java.util.List;
import java.util.Map;
import java.util.Objects;
import java.util.function.Function;
import java.util.stream.Collectors;

public class MainController {

    private static final Logger logger =
LoggerFactory.getLogger(MainController.class);

    private final MaterialCardDao materialCardDao;
    private final SupplierDao supplierDao;
    private final ToolDao toolDao;
    private final ProrabDao prorabDao;
    private final BuildingObjectDao buildingObjectDao;
    private final MaterialStockDao materialStockDao;
    private final MaterialReceiptDao materialReceiptDao;
    private final InvoiceDao invoiceDao;
    private final InvoiceItemDao invoiceItemDao;
    private final MaterialReturnDao materialReturnDao;
    private final ToolIssueDao toolIssueDao;

    private List<Map<String, String>> referenceRows = List.of();
    private List<Map<String, String>> warehouseRows = List.of();

```

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```

private List<Map<String, String>> reportRows = List.of();

@FXML
private TabPane referenceSubTabs;
@FXML
private TableView<Map<String, String>> referenceTableView;
@FXML
private TextField referenceSearchField;

@FXML
private TabPane warehouseSubTabs;
@FXML
private TableView<Map<String, String>> warehouseTableView;
@FXML
private TextField warehouseSearchField;

@FXML
private TabPane reportSubTabs;
@FXML
private TableView<Map<String, String>> reportTableView;
@FXML
private TextField reportSearchField;

public MainController(MaterialCardDao materialCardDao,
                      SupplierDao supplierDao,
                      ToolDao toolDao,
                      ProrabDao prorabDao,
                      BuildingObjectDao buildingObjectDao,
                      MaterialStockDao materialStockDao,
                      MaterialReceiptDao materialReceiptDao,
                      InvoiceDao invoiceDao,
                      InvoiceItemDao invoiceItemDao,
                      MaterialReturnDao materialReturnDao,
                      ToolIssueDao toolIssueDao) {
    this.materialCardDao = materialCardDao;
    this.supplierDao = supplierDao;
    this.toolDao = toolDao;
    this.prorabDao = prorabDao;
    this.buildingObjectDao = buildingObjectDao;
    this.materialStockDao = materialStockDao;
    this.materialReceiptDao = materialReceiptDao;
    this.invoiceDao = invoiceDao;
    this.invoiceItemDao = invoiceItemDao;
    this.materialReturnDao = materialReturnDao;
    this.toolIssueDao = toolIssueDao;
}

```

```

    }

    @FXML
    private void initialize() {

referenceSubTabs.getSelectionModel().selectedIndexProperty().addListener((obs,
oldVal, newVal) -> reloadReferenceView());

warehouseSubTabs.getSelectionModel().selectedIndexProperty().addListener((obs,
oldVal, newVal) -> reloadWarehouseView());
        reportSubTabs.getSelectionModel().selectedIndexProperty().addListener((obs,
oldVal, newVal) -> reloadReportView());
        refreshAll();
    }

    @FXML
    private void onRefreshReferenceClick() {
        reloadReferenceView();
    }

    @FXML
    private void onSearchReferenceClick() {
        applyFilter(referenceTableView, referenceRows, referenceSearchField);
    }

    @FXML
    private void onRefreshWarehouseClick() {
        reloadWarehouseView();
    }

    @FXML
    private void onSearchWarehouseClick() {
        applyFilter(warehouseTableView, warehouseRows, warehouseSearchField);
    }

    @FXML
    private void onRefreshReportClick() {
        reloadReportView();
    }

    @FXML
    private void onSearchReportClick() {
        applyFilter(reportTableView, reportRows, reportSearchField);
    }

```

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```

@FXML
private void onPrihodClick(ActionEvent event) {
    openDialog("prihod-view.fxml", "dialog.prihod", event,
PrihodController.class);
}

@FXML
private void onGiveMaterialClick(ActionEvent event) {
    openDialog("give-material-view.fxml", "dialog.giveMaterial", event,
GiveMaterialController.class);
}

@FXML
private void onGiveToolClick(ActionEvent event) {
    openDialog("give-tool-view.fxml", "dialog.giveTool", event,
GiveToolController.class);
}

@FXML
private void onReturnToolClick(ActionEvent event) {
    openDialog("return-tool-view.fxml", "dialog.returnTool", event,
ReturnToolController.class);
}

@FXML
private void onManageReferencesClick(ActionEvent event) {
    openDialog("reference-editor-view.fxml", "dialog.references", event,
ReferenceEditorController.class);
}

private void refreshAll() {
    reloadReferenceView();
    reloadWarehouseView();
    reloadReportView();
}

private void reloadReferenceView() {
    try {
        switch
(Math.max(referenceSubTabs.getSelectionModel().getSelectedIndex(), 0)) {
            case 1 -> loadSuppliersTable();
            case 2 -> loadToolsTable();
            case 3 -> loadProrabsTable();
            case 4 -> loadObjectsTable();
            default -> loadMaterialsTable();

```

```

    }
    applyFilter(referenceTableView, referenceRows, referenceSearchField);
} catch (SQLException e) {
    showError(e.getMessage());
}
}

private void reloadWarehouseView() {
    try {
        switch
(Math.max(warehouseSubTabs.getSelectionModel().getSelectedIndex(), 0)) {
            case 1 -> loadReceiptsTable();
            case 2 -> loadInvoicesTable();
            case 3 -> loadInvoiceItemsTable();
            case 4 -> loadReturnsTable();
            default -> loadStockTable();
        }
        applyFilter(warehouseTableView, warehouseRows, warehouseSearchField);
    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

private void reloadReportView() {
    try {
        switch (Math.max(reportSubTabs.getSelectionModel().getSelectedIndex(),
0)) {
            case 1 -> loadReportIssuesTable();
            case 2 -> loadReportDebtorsTable();
            case 3 -> loadReportToolIssuesTable();
            default -> loadReportReceiptsTable();
        }
        applyFilter(reportTableView, reportRows, reportSearchField);
    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

private void loadMaterialsTable() throws SQLException {
    List<String> keys = List.of("id", "name", "article", "unit",
"writeOffRate");
    List<String> headers = List.of("col.id", "label.name", "label.article",
"label.unit", "label.writeOffRate");
    List<Map<String, String>> rows = new ArrayList<>();
    for (MaterialCard card : materialCardDao.findAll()) {

```

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```

        rows.add(dataRow(keys,
            str(card.getId()), str(card.getName()), str(card.getArticle()),
            str(card.getUnit()), str(card.getWriteOffRate())));
    }
    showTable(referenceTableView, keys, headers, rows);
    referenceRows = rows;
}

private void loadSuppliersTable() throws SQLException {
    List<String> keys = List.of("id", "name", "contactPerson", "phone",
"email", "address");
    List<String> headers = List.of("col.id", "label.name",
"label.contactPerson", "label.phone", "label.email", "label.address");
    List<Map<String, String>> rows = new ArrayList<>();
    for (Supplier supplier : supplierDao.findAll()) {
        rows.add(dataRow(keys,
            str(supplier.getId()), str(supplier.getName()),
str(supplier.getContactPerson()),
            str(supplier.getPhone()), str(supplier.getEmail()),
str(supplier.getAddress())));
    }
    showTable(referenceTableView, keys, headers, rows);
    referenceRows = rows;
}

private void loadToolsTable() throws SQLException {
    List<String> keys = List.of("id", "name", "inventoryNumber", "condition");
    List<String> headers = List.of("col.id", "label.name",
"label.inventoryNumber", "label.condition");
    List<Map<String, String>> rows = new ArrayList<>();
    for (Tool tool : toolDao.findAll()) {
        rows.add(dataRow(keys,
            str(tool.getId()), str(tool.getName()),
str(tool.getInventoryNumber()), str(tool.getCondition())));
    }
    showTable(referenceTableView, keys, headers, rows);
    referenceRows = rows;
}

private void loadProrabsTable() throws SQLException {
    List<String> keys = List.of("id", "name", "phone");
    List<String> headers = List.of("col.id", "label.name", "label.phone");
    List<Map<String, String>> rows = new ArrayList<>();
    for (Prorab prorab : prorabDao.findAll()) {

```

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```

        rows.add(dataRow(keys, str(prorab.getId()), str(prorab.getName()),
str(prorab.getPhone())));
    }

    showTable(referenceTableView, keys, headers, rows);
    referenceRows = rows;
}

private void loadObjectsTable() throws SQLException {
    Map<Long, String> prorabNames = nameMap(prorabDao.findAll(), Prorab::getId,
Prorab::getName);
    List<String> keys = List.of("id", "name", "address", "prorab");
    List<String> headers = List.of("col.id", "label.name", "label.address",
"label.prorab");
    List<Map<String, String>> rows = new ArrayList<>();
    for (BuildingObject object : buildingObjectDao.findAll()) {
        String prorab = object.getProrabId() == null ? "" :
prorabNames.getDefault(object.getProrabId(), "?");
        rows.add(dataRow(keys, str(object.getId()), str(object.getName()),
str(object.getAddress()), prorab));
    }
    showTable(referenceTableView, keys, headers, rows);
    referenceRows = rows;
}

private void loadStockTable() throws SQLException {
    Map<Long, String> materialNames = nameMap(materialCardDao.findAll(),
MaterialCard::getId, MaterialCard::getName);
    List<String> keys = List.of("id", "material", "quantity", "lastUpdated");
    List<String> headers = List.of("col.id", "label.material",
"label.quantity", "col.lastUpdated");
    List<Map<String, String>> rows = new ArrayList<>();
    for (MaterialStock stock : materialStockDao.findAll()) {
        rows.add(dataRow(keys,
            str(stock.getId()),
            materialNames.getDefault(stock.getMaterialId(), "?"),
            str(stock.getQuantity()),
            str(stock.getLastUpdated())));
    }
    showTable(warehouseTableView, keys, headers, rows);
    warehouseRows = rows;
}

private void loadReceiptsTable() throws SQLException {
    Map<Long, String> materialNames = nameMap(materialCardDao.findAll(),
MaterialCard::getId, MaterialCard::getName);

```



```

        Map<Long, String> supplierNames = nameMap(supplierDao.findAll(),
Supplier::getId, Supplier::getName);

        List<String> keys = List.of("id", "material", "supplier", "quantity",
"price", "receiptDate");

        List<String> headers = List.of("col.id", "label.material",
"label.supplier", "label.quantity", "label.price", "label.receiptDate");

        List<Map<String, String>> rows = new ArrayList<>();
        for (MaterialReceipt receipt : materialReceiptDao.findAll()) {
            rows.add(dataRow(keys,
                str(receipt.getId()),
                materialNames.getOrDefault(receipt.getMaterialId(), "?"),
                supplierNames.getOrDefault(receipt.getSupplierId(), "?"),
                str(receipt.getQuantity()),
                str(receipt.getPrice()),
                str(receipt.getReceiptDate())));
        }

        showTable(warehouseTableView, keys, headers, rows);
        warehouseRows = rows;
    }

    private void loadInvoicesTable() throws SQLException {
        Map<Long, String> objectNames = nameMap(buildingObjectDao.findAll(),
BuildingObject::getId, BuildingObject::getName);

        List<String> keys = List.of("id", "invoiceNumber", "object", "issueDate");
        List<String> headers = List.of("col.id", "col.invoiceNumber",
"label.object", "label.issueDate");

        List<Map<String, String>> rows = new ArrayList<>();
        for (Invoice invoice : invoiceDao.findAll()) {
            rows.add(dataRow(keys,
                str(invoice.getId()),
                str(invoice.getInvoiceNumber()),
                objectNames.getOrDefault(invoice.getObjectId(), "?"),
                str(invoice.getIssueDate())));
        }

        showTable(warehouseTableView, keys, headers, rows);
        warehouseRows = rows;
    }

    private void loadInvoiceItemsTable() throws SQLException {
        Map<Long, String> materialNames = nameMap(materialCardDao.findAll(),
MaterialCard::getId, MaterialCard::getName);

        Map<Long, String> invoiceNumbers = invoiceNumberMap();

        List<String> keys = List.of("id", "invoiceNumber", "material", "quantity",
"priceAtMoment");

```

```

        List<String> headers = List.of("col.id", "col.invoiceNumber",
"label.material", "label.quantity", "label.priceAtIssue");
        List<Map<String, String>> rows = new ArrayList<>();
        for (InvoiceItem item : invoiceItemDao.findAll()) {
            rows.add(dataRow(keys,
                str(item.getId()),
                invoiceNumbers.getDefault(item.getInvoiceId(), "?"),
                materialNames.getDefault(item.getMaterialId(), "?"),
                str(item.getQuantity()),
                str(item.getPriceAtMoment())));
        }
        showTable(warehouseTableView, keys, headers, rows);
        warehouseRows = rows;
    }

    private void loadReturnsTable() throws SQLException {
        Map<Long, String> materialNames = nameMap(materialCardDao.findAll(),
MaterialCard::getId, MaterialCard::getName);
        Map<Long, String> invoiceNumbers = invoiceNumberMap();
        List<String> keys = List.of("id", "invoiceNumber", "material", "quantity",
"returnDate");
        List<String> headers = List.of("col.id", "col.invoiceNumber",
"label.material", "label.quantity", "label.materialReturnDate");
        List<Map<String, String>> rows = new ArrayList<>();
        for (MaterialReturn materialReturn : materialReturnDao.findAll()) {
            rows.add(dataRow(keys,
                str(materialReturn.getId()),
                invoiceNumbers.getDefault(materialReturn.getInvoiceId(),
"?"),
                materialNames.getDefault(materialReturn.getMaterialId(),
"?"),
                str(materialReturn.getQuantity()),
                str(materialReturn.getReturnDate())));
        }
        showTable(warehouseTableView, keys, headers, rows);
        warehouseRows = rows;
    }

    private void loadReportReceiptsTable() throws SQLException {
        loadReceiptsTable();
        reportRows = new ArrayList<>(warehouseRows);
        showTable(reportTableView,
            List.of("id", "material", "supplier", "quantity", "price",
"receiptDate"),

```

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```

        List.of("col.id", "label.material", "label.supplier",
"label.quantity", "label.price", "label.receiptDate"),
        reportRows);
    }

    private void loadReportIssuesTable() throws SQLException {
        loadInvoiceItemsTable();
        reportRows = new ArrayList<>(warehouseRows);
        showTable(reportTableView,
            List.of("id", "invoiceNumber", "material", "quantity",
"priceAtMoment"),
            List.of("col.id", "col.invoiceNumber", "label.material",
"label.quantity", "label.priceAtIssue"),
            reportRows);
    }

    private void loadReportDebtorsTable() throws SQLException {
        Map<Long, String> toolNames = nameMap(toolDao.findAll(), Tool::getId,
Tool::getName);
        Map<Long, String> objectNames = nameMap(buildingObjectDao.findAll(),
BuildingObject::getId, BuildingObject::getName);
        List<String> keys = List.of("id", "tool", "object", "issuedTo",
"issueDate", "returnDate");
        List<String> headers = List.of("col.id", "label.tool", "label.object",
"label.issuedTo", "label.issueDate", "label.returnDate");
        List<Map<String, String>> rows = new ArrayList<>();
        for (ToolIssue issue : toolIssueDao.findAll()) {
            if (issue.getActualReturnDate() != null) {
                continue;
            }
            rows.add(dataRow(keys,
                str(issue.getId()),
                toolNames.getOrDefault(issue.getToolId(), "?"),
                objectNames.getOrDefault(issue.getObjectId(), "?"),
                str(issue.getIssuedTo()),
                str(issue.getIssueDate()),
                str(issue.getReturnDate())));
        }
        showTable(reportTableView, keys, headers, rows);
        reportRows = rows;
    }

    private void loadReportToolIssuesTable() throws SQLException {
        Map<Long, String> toolNames = nameMap(toolDao.findAll(), Tool::getId,
Tool::getName);

```

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```

        Map<Long, String> objectNames = nameMap(buildingObjectDao.findAll(),
BuildingObject::getId, BuildingObject::getName);

        List<String> keys = List.of("id", "tool", "object", "issuedTo",
"issueDate", "returnDate", "actualReturnDate");

        List<String> headers = List.of("col.id", "label.tool", "label.object",
"label.issuedTo", "label.issueDate", "label.returnDate", "label.actualReturnDate");

        List<Map<String, String>> rows = new ArrayList<>();
        for (ToolIssue issue : toolIssueDao.findAll()) {
            rows.add(dataRow(keys,
                str(issue.getId()),
                toolNames.getOrDefault(issue.getToolId(), "?"),
                objectNames.getOrDefault(issue.getObjectId(), "?"),
                str(issue.getIssuedTo()),
                str(issue.getIssueDate()),
                str(issue.getReturnDate()),
                str(issue.getActualReturnDate())));
        }
        showTable(reportTableView, keys, headers, rows);
        reportRows = rows;
    }

    private Map<Long, String> invoiceNumberMap() throws SQLException {
        Map<Long, String> invoiceNumbers = new HashMap<>();
        for (Invoice invoice : invoiceDao.findAll()) {
            invoiceNumbers.put(invoice.getId(), invoice.getInvoiceNumber());
        }
        return invoiceNumbers;
    }

    private void showTable(Table<Map<String, String>> table,
        List<String> keys,
        List<String> headerKeys,
        List<Map<String, String>> rows) {
        table.getColumns().clear();
        for (int i = 0; i < keys.size(); i++) {
            String key = keys.get(i);
            TableColumn<Map<String, String>, String> column =
                new
TableColumn<>(Messages.bundle.getString(headerKeys.get(i)));
            column.setCellValueFactory(data -> new
SimpleStringProperty(data.getValue().getOrDefault(key, "")));
            table.getColumns().add(column);
        }
        table.setItems(FXCollections.observableArrayList(rows));
    }
}

```

```

private void applyFilter(Table<Map<String, String>> table,
                        List<Map<String, String>> allRows,
                        TextField searchField) {
    String filter = searchField.getText().trim().toLowerCase();
    if (filter.isEmpty()) {
        table.setItems(FXCollections.observableArrayList(allRows));
        return;
    }
    table.setItems(FXCollections.observableArrayList(
        allRows.stream()
            .filter(row -> rowMatches(row, filter))
            .collect(Collectors.toList())
    ));
}

private static boolean rowMatches(Map<String, String> row, String filter) {
    return row.values().stream()
        .filter(Objects::nonNull)
        .map(String::toLowerCase)
        .anyMatch(value -> value.contains(filter));
}

private static Map<String, String> dataRow(List<String> keys, String... values)
{
    Map<String, String> row = new LinkedHashMap<>();
    for (int i = 0; i < keys.size(); i++) {
        row.put(keys.get(i), i < values.length && values[i] != null ? values[i]
: "");
    }
    return row;
}

private static <T> Map<Long, String> nameMap(List<T> items, Function<T, Long>
idFn, Function<T, String> nameFn) {
    Map<Long, String> map = new HashMap<>();
    for (T item : items) {
        map.put(idFn.apply(item), nameFn.apply(item));
    }
    return map;
}

private static String str(Object value) {
    return value == null ? "" : String.valueOf(value);
}

```

```

        private void openDialog(String fxml, String titleKey, ActionEvent event,
Class<?> controllerClass) {
            logger.info("Открытие окна: {}", titleKey);
            try {
                FXMLLoader loader = new
FXMLLoader(getClass().getResource("/ru/kochedykovdev/laba7/" + fxml),
Messages.bundle);
                loader.setControllerFactory(clazz -> createDialogController(clazz,
controllerClass));

                Scene scene = new Scene(loader.load());
                Stage stage = new Stage();
                stage.setTitle(Messages.bundle.getString(titleKey));
                stage.setScene(scene);
                stage.initModality(Modality.WINDOW_MODAL);
                stage.initOwner(((Stage) ((Node)
event.getSource()).getScene().getWindow()));
                stage.showAndWait();

                refreshAll();
            } catch (IOException e) {
                showError(e.getMessage());
            }
        }

        private Object createDialogController(Class<?> clazz, Class<?> expected) {
            if (clazz != expected) {
                throw new IllegalStateException("Неизвестный контроллер: " + clazz);
            }
            if (clazz == PrihodController.class) {
                return new PrihodController(materialCardDao, supplierDao,
materialReceiptDao, materialStockDao);
            }
            if (clazz == GiveMaterialController.class) {
                return new GiveMaterialController(invoiceDao, invoiceItemDao,
buildingObjectDao, materialCardDao, materialStockDao);
            }
            if (clazz == GiveToolController.class) {
                return new GiveToolController(toolDao, buildingObjectDao,
toolIssueDao);
            }
            if (clazz == ReturnToolController.class) {
                return new ReturnToolController(toolIssueDao, toolDao);
            }
        }

```

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```

        if (clazz == ReferenceEditorController.class) {
            return new ReferenceEditorController(
                materialCardDao, supplierDao, prorabDao, buildingObjectDao,
                toolDao
            );
        }
        throw new IllegalStateException("Неизвестный контроллер: " + clazz);
    }

    private void showError(String message) {
        logger.error("{} ", message);
        Alert alert = new Alert(Alert.AlertType.ERROR);
        alert.setHeaderText(Messages.bundle.getString("error.header"));
        alert.setContentText(message);
        alert.showAndWait();
    }
}

```

Листинг 4. PrihodController.java

```

package ru.kochedykovdev.laba7.controllers;

import javafx.beans.binding.Bindings;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.scene.Node;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.DatePicker;
import javafx.scene.control.ListCell;
import javafx.scene.control.TextField;
import javafx.stage.Stage;
import ru.kochedykovdev.laba7.dao.MaterialCardDao;
import ru.kochedykovdev.laba7.dao.MaterialReceiptDao;
import ru.kochedykovdev.laba7.dao.MaterialStockDao;
import ru.kochedykovdev.laba7.dao.SupplierDao;
import ru.kochedykovdev.laba7.model.MaterialCard;
import ru.kochedykovdev.laba7.model.MaterialReceipt;
import ru.kochedykovdev.laba7.model.MaterialStock;
import ru.kochedykovdev.laba7.model.Supplier;
import ru.kochedykovdev.laba7.util.Messages;
import ru.kochedykovdev.laba7.util.FieldValidation;

import java.math.BigDecimal;

```

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```

import java.sql.SQLException;
import java.time.LocalDateTime;
import java.util.Optional;

public class PrihodController {

    private final MaterialCardDao materialCardDao;
    private final SupplierDao supplierDao;
    private final MaterialReceiptDao materialReceiptDao;
    private final MaterialStockDao materialStockDao;

    @FXML
    private ComboBox<MaterialCard> materialComboBox;
    @FXML
    private ComboBox<Supplier> supplierComboBox;
    @FXML
    private TextField quantityField;
    @FXML
    private TextField priceField;
    @FXML
    private DatePicker receiptDatePicker;
    @FXML
    private Button submitButton;

    public PrihodController(MaterialCardDao materialCardDao,
                           SupplierDao supplierDao,
                           MaterialReceiptDao materialReceiptDao,
                           MaterialStockDao materialStockDao) {

        this.materialCardDao = materialCardDao;
        this.supplierDao = supplierDao;
        this.materialReceiptDao = materialReceiptDao;
        this.materialStockDao = materialStockDao;
    }

    @FXML
    private void initialize() {
        try {
            setupCombo(materialComboBox);
            setupCombo(supplierComboBox);
            materialComboBox.getItems().setAll(materialCardDao.findAll());
            supplierComboBox.getItems().setAll(supplierDao.findAll());
            receiptDatePicker.setValue(java.time.LocalDate.now());

            FieldValidation.onlyDecimal(quantityField);
            FieldValidation.onlyDecimal(priceField);
        }
    }
}

```

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```

        FieldValidation.bindSubmit(submitButton, Bindings.createBooleanBinding(
            () -> materialComboBox.getValue() != null
                && supplierComboBox.getValue() != null
                && !quantityField.getText().isBlank()
                && !priceField.getText().isBlank()
                && receiptDatePicker.getValue() != null
                && isPositive(quantityField.getText())
                && isPositive(priceField.getText()),
            materialComboBox.valueProperty(),
            supplierComboBox.valueProperty(),
            quantityField.textProperty(),
            priceField.textProperty(),
            receiptDatePicker.valueProperty()
        ));
    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

@FXML
private void onCloseClick(ActionEvent event) {
    closeWindow(event);
}

@FXML
private void onSubmitClick(ActionEvent event) {
    try {
        MaterialCard material = materialComboBox.getValue();
        BigDecimal qty = new BigDecimal(quantityField.getText().trim());

        MaterialReceipt receipt = new MaterialReceipt(
            null,
            material.getId(),
            supplierComboBox.getValue().getId(),
            qty,
            new BigDecimal(priceField.getText().trim()),
            receiptDatePicker.getValue()
        );

        materialReceiptDao.insert(receipt);

        updateStock(material.getId(), qty);
        closeWindow(event);
    } catch (Exception e) {
        showError(e.getMessage());
    }
}

```

```

    }

    }

    private void updateStock(Long materialId, BigDecimal qty) throws SQLException {
        Optional<MaterialStock> existing = materialStockDao.findAll().stream()
            .filter(s -> materialId.equals(s.getMaterialId()))
            .findFirst();

        if (existing.isPresent()) {
            MaterialStock stock = existing.get();
            stock.setQuantity(stock.getQuantity().add(qty));
            stock.setLastUpdated(LocalDateTime.now());
            materialStockDao.update(stock);
        } else {
            MaterialStock stock = new MaterialStock(null, materialId, qty,
LocalDateTime.now());
            materialStockDao.insert(stock);
        }
    }

    private static boolean isPositive(String text) {
        if (text.isBlank()) {
            return false;
        }
        return new BigDecimal(text.trim()).compareTo(BigDecimal.ZERO) > 0;
    }

    private static <T> void setupCombo(ComboBox<T> comboBox) {
        comboBox.setCellFactory(list -> new ListCell<>() {
            @Override
            protected void updateItem(T item, boolean empty) {
                super.updateItem(item, empty);
                setText(empty || item == null ? null : item.toString());
            }
        });
        comboBox.setButtonCell(comboBox.getCellFactory().call(null));
    }

    private void closeWindow(ActionEvent event) {
        Stage stage = (Stage) ((Node) event.getSource()).getScene().getWindow();
        stage.close();
    }

    private void showError(String message) {
        Alert alert = new Alert(Alert.AlertType.ERROR);

```

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```

        alert.setHeaderText(Messages.bundle.getString("error.header"));
        alert.setContentText(message);
        alert.showAndWait();
    }
}

```

Листинг 5. ReferenceEditorController.java

```

package ru.kochedykovdev.laba7.controllers;

import javafx.beans.binding.Bindings;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.scene.Node;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.ListCell;
import javafx.scene.control.Tab;
import javafx.scene.control.TabPane;
import javafx.scene.control.TextField;
import javafx.stage.Stage;
import ru.kochedykovdev.laba7.dao.BuildingObjectDao;
import ru.kochedykovdev.laba7.dao.MaterialCardDao;
import ru.kochedykovdev.laba7.dao.ProrabDao;
import ru.kochedykovdev.laba7.dao.SupplierDao;
import ru.kochedykovdev.laba7.dao.ToolDao;
import ru.kochedykovdev.laba7.model.BuildingObject;
import ru.kochedykovdev.laba7.model.MaterialCard;
import ru.kochedykovdev.laba7.model.Prorab;
import ru.kochedykovdev.laba7.model.Supplier;
import ru.kochedykovdev.laba7.model.Tool;
import ru.kochedykovdev.laba7.util.FieldValidation;
import ru.kochedykovdev.laba7.util.Messages;

import java.sql.SQLException;

public class ReferenceEditorController {

    private final MaterialCardDao materialCardDao;
    private final SupplierDao supplierDao;
    private final ProrabDao prorabDao;
    private final BuildingObjectDao buildingObjectDao;
    private final ToolDao toolDao;

```

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```

@FXML
private TabPane referenceTabPane;

@FXML
private Tab objectTab;


@FXML
private TextField materialNameField;
@FXML
private TextField materialArticleField;
@FXML
private TextField materialUnitField;
@FXML
private TextField materialWriteOffRateField;
@FXML
private Button materialSubmitButton;


@FXML
private TextField supplierNameField;
@FXML
private TextField supplierContactField;
@FXML
private TextField supplierPhoneField;
@FXML
private TextField supplierEmailField;
@FXML
private TextField supplierAddressField;
@FXML
private Button supplierSubmitButton;


@FXML
private TextField prorabNameField;
@FXML
private TextField prorabPhoneField;
@FXML
private Button prorabSubmitButton;


@FXML
private TextField objectNameField;
@FXML
private TextField objectAddressField;
@FXML
private ComboBox<Prorab> objectProrabComboBox;
@FXML
private Button objectSubmitButton;

```

```

@FXML
private TextField toolNameField;

@FXML
private TextField toolInventoryField;

@FXML
private TextField toolConditionField;

@FXML
private Button toolSubmitButton;

public ReferenceEditorController(MaterialCardDao materialCardDao,
                                SupplierDao supplierDao,
                                ProrabDao prorabDao,
                                BuildingObjectDao buildingObjectDao,
                                ToolDao toolDao) {

    this.materialCardDao = materialCardDao;
    this.supplierDao = supplierDao;
    this.prorabDao = prorabDao;
    this.buildingObjectDao = buildingObjectDao;
    this.toolDao = toolDao;
}

@FXML
private void initialize() {
    FieldValidation.onlyInteger(materialArticleField);
    FieldValidation.onlyInteger(materialWriteOffRateField);
    setupCombo(objectProrabComboBox);

    FieldValidation.bindSubmit(materialSubmitButton,
Bindings.createBooleanBinding(
        () -> !materialNameField.getText().isBlank(),
        materialNameField.textProperty()
    ));
    FieldValidation.bindSubmit(supplierSubmitButton,
Bindings.createBooleanBinding(
        () -> !supplierNameField.getText().isBlank(),
        supplierNameField.textProperty()
    ));
    FieldValidation.bindSubmit(prorabSubmitButton,
Bindings.createBooleanBinding(
        () -> !prorabNameField.getText().isBlank(),
        prorabNameField.textProperty()
    ));
    FieldValidation.bindSubmit(objectSubmitButton,
Bindings.createBooleanBinding(
        () -> !objectNameField.getText().isBlank(),

```

```

        objectNameField.textProperty()

    ));
    FieldValidation.bindSubmit(toolSubmitButton, Bindings.createBooleanBinding(
        () -> !toolNameField.getText().isBlank(),
        toolNameField.textProperty()
    ));

referenceTabPane.getSelectionModel().selectedItemProperty().addListener((obs,
oldTab, newTab) -> {
    if (newTab == objectTab) {
        reloadProrabs();
    }
});
reloadProrabs();
}

@FXML
private void onCloseClick(ActionEvent event) {
    closeWindow(event);
}

@FXML
private void onMaterialSubmitClick(ActionEvent event) {
    try {
        MaterialCard card = new MaterialCard(
            null,
            materialNameField.getText().trim(),
            parseOptionalInteger(materialArticleField),
            blankToNull(materialUnitField),
            parseOptionalInteger(materialWriteOffRateField)
        );
        materialCardDao.insert(card);
        clearMaterialForm();
        closeWindow(event);
    } catch (Exception e) {
        showError(e.getMessage());
    }
}

@FXML
private void onSupplierSubmitClick(ActionEvent event) {
    try {
        Supplier supplier = new Supplier(
            null,

```

```

        supplierNameField.getText().trim(),
        blankToNull(supplierContactField),
        blankToNull(supplierPhoneField),
        blankToNull(supplierEmailField),
        blankToNull(supplierAddressField)
    );
    supplierDao.insert(supplier);
    clearSupplierForm();
    closeWindow(event);
} catch (Exception e) {
    showError(e.getMessage());
}
}

@FXML
private void onProrabSubmitClick(ActionEvent event) {
    try {
        Prorab prorab = new Prorab(
            null,
            prorabNameField.getText().trim(),
            blankToNull(prorabPhoneField)
        );
        prorabDao.insert(prorab);
        clearProrabForm();
        reloadProrabs();
        closeWindow(event);
    } catch (Exception e) {
        showError(e.getMessage());
    }
}

@FXML
private void onObjectSubmitClick(ActionEvent event) {
    try {
        Prorab prorab = objectProrabComboBox.getValue();
        BuildingObject object = new BuildingObject(
            null,
            objectNameField.getText().trim(),
            blankToNull(objectAddressField),
            prorab != null ? prorab.getId() : null
        );
        buildingObjectDao.insert(object);
        clearObjectForm();
        closeWindow(event);
    } catch (Exception e) {

```

```

        showError(e.getMessage());
    }
}

@FXML
private void onToolSubmitClick(ActionEvent event) {
    try {
        Tool tool = new Tool(
            null,
            toolNameField.getText().trim(),
            blankToNull(toolInventoryField),
            blankToNull(toolConditionField)
        );
        toolDao.insert(tool);
        clearToolForm();
        closeWindow(event);
    } catch (Exception e) {
        showError(e.getMessage());
    }
}

private void reloadProrabs() {
    try {
        objectProrabComboBox.getItems().setAll(prorabDao.findAll());
    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

private void clearMaterialForm() {
    materialNameField.clear();
    materialArticleField.clear();
    materialUnitField.clear();
    materialWriteOffRateField.clear();
}

private void clearSupplierForm() {
    supplierNameField.clear();
    supplierContactField.clear();
    supplierPhoneField.clear();
    supplierEmailField.clear();
    supplierAddressField.clear();
}

private void clearProrabForm() {

```

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```

        prorabNameField.clear();
        prorabPhoneField.clear();
    }

    private void clearObjectForm() {
        objectNameField.clear();
        objectAddressField.clear();
        objectProrabComboBox.getSelectionModel().clearSelection();
    }

    private void clearToolForm() {
        toolNameField.clear();
        toolInventoryField.clear();
        toolConditionField.clear();
    }

    private static Integer parseOptionalInteger(TextField field) {
        String text = field.getText().trim();
        if (text.isEmpty()) {
            return null;
        }
        return Integer.valueOf(text);
    }

    private static String blankToNull(TextField field) {
        String text = field.getText().trim();
        return text.isEmpty() ? null : text;
    }

    private static <T> void setupCombo(ComboBox<T> comboBox) {
        comboBox.setCellFactory(list -> new ListCell<>() {
            @Override
            protected void updateItem(T item, boolean empty) {
                super.updateItem(item, empty);
                setText(empty || item == null ? null : item.toString());
            }
        });
        comboBox.setButtonCell(comboBox.getCellFactory().call(null));
    }

    private void closeWindow(ActionEvent event) {
        Stage stage = (Stage) ((Node) event.getSource()).getScene().getWindow();
        stage.close();
    }

```

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```

        private void showError(String message) {
            Alert alert = new Alert(Alert.AlertType.ERROR);
            alert.setHeaderText(Messages.bundle.getString("error.header"));
            alert.setContentText(message);
            alert.showAndWait();
        }
    }
}

```

Листинг 6. ReturnToolController.java

```

package ru.kochedykovdev.laba7.controllers;

import javafx.beans.binding.Bindings;
import javafx.event.ActionEvent;
import javafx.fxml.FXML;
import javafx.scene.Node;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.ComboBox;
import javafx.scene.control.DatePicker;
import javafx.scene.control.ListCell;
import javafx.scene.control.TextField;
import javafx.stage.Stage;
import ru.kochedykovdev.laba7.dao.ToolDao;
import ru.kochedykovdev.laba7.dao.ToolIssueDao;
import ru.kochedykovdev.laba7.model.Tool;
import ru.kochedykovdev.laba7.model.ToolIssue;
import ru.kochedykovdev.laba7.util.Messages;
import ru.kochedykovdev.laba7.util.FieldValidation;

import java.sql.SQLException;
import java.util.HashMap;
import java.util.Map;

public class ReturnToolController {

    private final ToolIssueDao toolIssueDao;
    private final ToolDao toolDao;

    private final Map<Long, String> toolNames = new HashMap<>();

    @FXML
    private ComboBox<ToolIssue> toolIssueComboBox;

    @FXML
    private DatePicker actualReturnDatePicker;

```

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```

@FXML
private TextField conditionField;

@FXML
private Button submitButton;

public ReturnToolController(ToolIssueDao toolIssueDao, ToolDao toolDao) {
    this.toolIssueDao = toolIssueDao;
    this.toolDao = toolDao;
}

@FXML
private void initialize() {
    try {
        for (Tool tool : toolDao.findAll()) {
            toolNames.put(tool.getId(), tool.getName());
        }

        toolIssueComboBox.setCellFactory(list -> new ListCell<>() {
            @Override
            protected void updateItem(ToolIssue item, boolean empty) {
                super.updateItem(item, empty);
                setText(empty || item == null ? null : formatIssue(item));
            }
        });

        toolIssueComboBox.setButtonCell(toolIssueComboBox.getCellFactory().call(null));

        toolIssueComboBox.getItems().setAll(
            toolIssueDao.findAll().stream()
                .filter(i -> i.getActualReturnDate() == null)
                .toList()
        );

        actualReturnDatePicker.setValue(java.time.LocalDate.now());

        FieldValidation.bindSubmit(submitButton, Bindings.createBooleanBinding(
            () -> toolIssueComboBox.getValue() != null
                && actualReturnDatePicker.getValue() != null
                && !conditionField.getText().isBlank(),
            toolIssueComboBox.valueProperty(),
            actualReturnDatePicker.valueProperty(),
            conditionField.textProperty()
        ));

    } catch (SQLException e) {
        showError(e.getMessage());
    }
}

```

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```

    }

    @FXML
    private void onCloseClick(ActionEvent event) {
        closeWindow(event);
    }

    @FXML
    private void onSubmitClick(ActionEvent event) {
        try {
            ToolIssue issue = toolIssueComboBox.getValue();
            issue.setActualReturnDate(actualReturnDatePicker.getValue());
            toolIssueDao.update(issue);

            Tool tool = toolDao.findById(issue.getToolId())
                .orElseThrow(() -> new IllegalStateException("Инструмент не
найден"));

            tool.setCondition(conditionField.getText().trim());
            toolDao.update(tool);

            closeWindow(event);
        } catch (Exception e) {
            showError(e.getMessage());
        }
    }

    private String formatIssue(ToolIssue issue) {
        String name = toolNames.getOrDefault(issue.getToolId(), "инструмент");
        return name + " - " + issue.getIssuedTo();
    }

    private void closeWindow(ActionEvent event) {
        Stage stage = (Stage) ((Node) event.getSource()).getScene().getWindow();
        stage.close();
    }

    private void showError(String message) {
        Alert alert = new Alert(Alert.AlertType.ERROR);
        alert.setHeaderText(Messages.bundle.getString("error.header"));
        alert.setContentText(message);
        alert.showAndWait();
    }
}

```

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Листинг 7. util/DBHelper.java

```
package ru.kochedykovdev.laba7.util;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import java.io.IOException;
import java.io.InputStream;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.util.Properties;

public class DBHelper {

    private static final String CONFIG_PATH =
"/ru/kochedykovdev/laba7/config.properties";

    private static final Logger logger = LoggerFactory.getLogger(DBHelper.class);

    private static final String dbUrlBase;
    private static final String dbName;
    public static final String SCHEMA;

    private static Connection connection;
    private static String storedUser;
    private static String storedPassword;

    static {
        Properties prop = new Properties();
        try (InputStream is = DBHelper.class.getResourceAsStream(CONFIG_PATH)) {
            if (is == null) {
                throw new IOException("config.properties не найден");
            }
            prop.load(is);
            dbUrlBase = prop.getProperty("db.url");
            dbName = prop.getProperty("db.name");
            SCHEMA = prop.getProperty("db.schema");
            logger.debug("Загружены настройки подключения (url, name, schema)");
        } catch (IOException ex) {
            logger.error("Ошибка загрузки config.properties", ex);
            throw new ExceptionInInitializerError(ex);
        }
    }
}
```

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```

    public static void initConnection(String user, String password) throws
SQLException {
        storedUser = user;
        storedPassword = password;
        connect(true);
    }

    public static Connection getConnection() throws SQLException {
        if (storedUser == null) {
            throw new SQLException("Соединение не инициализировано. Вызовите
initConnection()");
        }
        if (connection == null || connection.isClosed()) {
            connect(false);
        }
        return connection;
    }

    private static void connect(boolean logConnect) throws SQLException {
        if (connection != null && !connection.isClosed()) {
            closeConnection();
        }
        String fullUrl = dbUrlBase + dbName;
        if (logConnect) {
            logger.info("Подключение к {} пользователем {}", fullUrl, storedUser);
        } else {
            logger.debug("Повторное подключение к {} пользователем {}", fullUrl,
storedUser);
        }
        connection = DriverManager.getConnection(fullUrl, storedUser,
storedPassword);
        connection.setSchema(SCHEMA);
        logger.info("Соединение установлено, схема: {}", SCHEMA);
    }

    public static void closeConnection() {
        if (connection != null) {
            try {
                connection.close();
                logger.info("Соединение закрыто");
            } catch (SQLException ex) {
                logger.error("Ошибка закрытия соединения", ex);
            } finally {
                connection = null;
            }
        }
    }

```

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```

    }

    }

}

```

Листинг 8. util/LoginDialog.java

```

package ru.kochedykovdev.laba7.util;

import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.PasswordField;
import javafx.scene.control.TextField;
import javafx.scene.layout.GridPane;
import javafx.scene.layout.HBox;
import javafx.scene.layout.VBox;
import javafx.stage.Modality;
import javafx.stage.Stage;
import lombok.AllArgsConstructor;
import lombok.Getter;

import java.util.Optional;

public class LoginDialog {

    public Optional<LoginResult> showAndWait() {
        Stage dialogStage = new Stage();
        dialogStage.initModality(Modality.APPLICATION_MODAL);
        dialogStage.setTitle(Messages.bundle.getString("login.title"));

        GridPane grid = new GridPane();
        grid.setPadding(new Insets(10));
        grid.setHgap(10);
        grid.setVgap(10);

        Label userLabel = new Label(Messages.bundle.getString("login.username"));
        TextField userField = new TextField();
        Label passLabel = new Label(Messages.bundle.getString("login.password"));
        PasswordField passField = new PasswordField();

        grid.add(userLabel, 0, 0);
        grid.add(userField, 1, 0);
        grid.add(passLabel, 0, 1);
        grid.add(passField, 1, 1);

        Button btnOk = new Button(Messages.bundle.getString("login.submit"));
        Button btnCancel = new Button(Messages.bundle.getString("login.cancel"));
        HBox buttonBar = new HBox(10, btnOk, btnCancel);
        buttonBar.setAlignment(Pos.CENTER_RIGHT);

        VBox root = new VBox(15, grid, buttonBar);
        root.setPadding(new Insets(10));

        final String[] username = {null};
        final String[] password = {null};
        final boolean[] okClicked = {false};

        btnOk.setOnAction(e -> {
            String u = userField.getText().trim();
            String p = passField.getText();

```

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```

        if (u.isEmpty()) {
            showAlert(Alert.AlertType.ERROR,
Messages.bundle.getString("error.header"),
                Messages.bundle.getString("login.error.emptyUsername"));
            return;
        }
        username[0] = u;
        password[0] = p;
        okClicked[0] = true;
        dialogStage.close();
    });

    btnCancel.setOnAction(e -> {
        okClicked[0] = false;
        dialogStage.close();
    });

    dialogStage.setScene(new Scene(root, 320, 160));
    dialogStage.showAndWait();

    if (okClicked[0]) {
        return Optional.of(new LoginResult(username[0], password[0]));
    }
    return Optional.empty();
}

private void showAlert(Alert.AlertType type, String title, String message) {
    Alert alert = new Alert(type);
    alert.setTitle(title);
    alert.setHeaderText(null);
    alert.setContentText(message);
    alert.showAndWait();
}

@Getter
@AllArgsConstructor
public static class LoginResult {
    private final String username;
    private final String password;
}
}

```

Листинг 9. GiveMaterialController.java

```

package ru.kochedykovdev.laba7.util;

import java.util.Locale;
import java.util.ResourceBundle;

public class Messages {

    public static final ResourceBundle bundle =
        //ResourceBundle.getBundle("ru.kochedykovdev.laba7.messages",
Locale.ENGLISH);

        //ResourceBundle.getBundle("ru.kochedykovdev.laba7.messages",
Locale.forLanguageTag("ru"));

        //ResourceBundle.getBundle("ru.kochedykovdev.laba7.messages",
Locale.forLanguageTag("ua"));
}

```

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```

        ResourceBundle.getBundle("ru.kochedykovdev.laba7.messages",
Locale.getDefault());
}

```

Листинг 10. HelloApplication.java

```

package ru.kochedykovdev.laba7;

import javafx.application.Application;
import javafx.application.Platform;
import javafx.fxml.FXMLLoader;
import javafx.scene.Scene;
import javafx.scene.control.Alert;
import javafx.stage.Stage;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import ru.kochedykovdev.laba7.controllers.MainController;
import ru.kochedykovdev.laba7.dao.BuildingObjectDao;
import ru.kochedykovdev.laba7.dao.InvoiceDao;
import ru.kochedykovdev.laba7.dao.InvoiceItemDao;
import ru.kochedykovdev.laba7.dao.MaterialCardDao;
import ru.kochedykovdev.laba7.dao.MaterialReceiptDao;
import ru.kochedykovdev.laba7.dao.MaterialReturnDao;
import ru.kochedykovdev.laba7.dao.MaterialStockDao;
import ru.kochedykovdev.laba7.dao.ProrabDao;
import ru.kochedykovdev.laba7.dao.SupplierDao;
import ru.kochedykovdev.laba7.dao.ToolDao;
import ru.kochedykovdev.laba7.dao.ToolIssueDao;
import ru.kochedykovdev.laba7.dao.impl.BuildingObjectDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.InvoiceDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.InvoiceItemDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.MaterialCardDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.MaterialReceiptDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.MaterialReturnDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.MaterialStockDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.ProrabDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.SupplierDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.ToolDaoImpl;
import ru.kochedykovdev.laba7.dao.impl.ToolIssueDaoImpl;
import ru.kochedykovdev.laba7.util.DBHelper;
import ru.kochedykovdev.laba7.util.LoginDialog;
import ru.kochedykovdev.laba7.util.Messages;

import java.io.IOException;
import java.sql.SQLException;

```

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```

import java.util.Locale;
import java.util.Optional;

public class HelloApplication extends Application {

    private static final Logger logger =
LoggerFactory.getLogger(HelloApplication.class);

    @Override
    public void start(Stage stage) {
        logger.info("Приложение запущено");
        logger.info("Локаль: {}, ресурсы: {}", Locale.getDefault(),
Messages.bundle.getLocale());

        LoginDialog loginDialog = new LoginDialog();
        while (true) {
            Optional<LoginDialog.LoginResult> result = loginDialog.showAndWait();
            if (result.isEmpty()) {
                logger.info("Пользователь отменил вход, выход");
                Platform.exit();
                return;
            }

            String username = result.get().getUsername();
            String password = result.get().getPassword();
            try {
                DBHelper.initConnection(username, password);
                logger.info("Успешное подключение для {}", username);
                break;
            } catch (SQLException ex) {
                logger.error("Ошибка подключения для {}: {}", username,
ex.getMessage());
                Alert alert = new Alert(Alert.AlertType.ERROR);

                alert.setTitle(Messages.bundle.getString("login.error.connectionTitle"));

                alert.setHeaderText(Messages.bundle.getString("login.error.connectionHeader"));

                alert.setContentText(Messages.bundle.getString("login.error.connectionMessage")
+ "\n\n" + ex.getMessage());

                alert.showAndWait();
            }
        }

        try {

```

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```

MaterialCardDao materialCardDao = new MaterialCardDaoImpl();
SupplierDao supplierDao = new SupplierDaoImpl();
ToolDao toolDao = new ToolDaoImpl();
ProrabDao prorabDao = new ProrabDaoImpl();
BuildingObjectDao buildingObjectDao = new BuildingObjectDaoImpl();
MaterialStockDao materialStockDao = new MaterialStockDaoImpl();
MaterialReceiptDao materialReceiptDao = new MaterialReceiptDaoImpl();
InvoiceDao invoiceDao = new InvoiceDaoImpl();
InvoiceItemDao invoiceItemDao = new InvoiceItemDaoImpl();
MaterialReturnDao materialReturnDao = new MaterialReturnDaoImpl();
ToolIssueDao toolIssueDao = new ToolIssueDaoImpl();

FXMLLoader loader = new
FXMLLoader(HelloApplication.class.getResource("main-view.fxml"), Messages.bundle);
    loader.setControllerFactory(clazz -> {
        if (clazz == MainController.class) {
            return new MainController(
                materialCardDao, supplierDao, toolDao, prorabDao,
buildingObjectDao,
                materialStockDao, materialReceiptDao, invoiceDao,
invoiceItemDao,
                materialReturnDao, toolIssueDao
            );
        }
        throw new IllegalStateException("Неизвестный контроллер: " +
clazz);
    });

Scene scene = new Scene(loader.load(), 1000, 600);
stage.setTitle(Messages.bundle.getString("app.title"));
stage.setScene(scene);
stage.show();
logger.info("Главное окно открыто");
} catch (IOException e) {
    logger.error("Ошибка загрузки FXML", e);
    Platform.exit();
}
}

@Override
public void stop() {
    DBHelper.closeConnection();
    logger.info("Приложение завершено");
}

```

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```
}
```

Листинг 11. give-material-view.fxml

```
<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.Cursor?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.ComboBox?>
<?import javafx.scene.control.DatePicker?>
<?import javafx.scene.control.Label?>
<?import javafx.scene.control.Separator?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.ColumnConstraints?>
<?import javafx.scene.layout.GridPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.layout.RowConstraints?>
<?import javafx.scene.layout.VBox?>

<VBox spacing="15.0" xmlns="http://javafx.com/javafx/25"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.GiveMaterialController">
    <children>
        <GridPane hgap="10.0" vgap="12.0">
            <columnConstraints>
                <ColumnConstraints minWidth="120.0" prefWidth="140.0" />
                <ColumnConstraints hgrow="ALWAYS" minWidth="200.0" />
            </columnConstraints>
            <children>
                <Label text="%label.invoiceNo" GridPane.columnIndex="0"
GridPane.rowIndex="0" />
                <TextField fx:id="invoiceNumberField" maxWidth="1.7976931348623157E308"
promptText="%prompt.invoiceExample" GridPane.columnIndex="1" GridPane.rowIndex="0"
/>
                <Label text="%label.object" GridPane.columnIndex="0"
GridPane.rowIndex="1" />
                <ComboBox fx:id="objectComboBox" maxWidth="1.7976931348623157E308"
promptText="%prompt.selectObject" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                <Label text="%label.issueDate" GridPane.columnIndex="0"
GridPane.rowIndex="2" />
                <DatePicker fx:id="issueDatePicker" GridPane.columnIndex="1"
GridPane.rowIndex="2" />
                <Label text="%label.material" GridPane.columnIndex="0"
GridPane.rowIndex="3" />
            </children>
        </GridPane>
    </children>
</VBox>
```

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```

        <ComboBox fx:id="materialComboBox" maxWidth="1.7976931348623157E308"
promptText="%prompt.selectMaterial" GridPane.columnIndex="1" GridPane.rowIndex="3"
/>

        <Label text="%label.quantity" GridPane.columnIndex="0"
GridPane.rowIndex="4" />

        <TextField fx:id="quantityField" maxWidth="1.7976931348623157E308"
promptText="0" GridPane.columnIndex="1" GridPane.rowIndex="4" />

        <Label text="%label.priceAtIssue" GridPane.columnIndex="0"
GridPane.rowIndex="5" />

        <TextField fx:id="priceField" maxWidth="1.7976931348623157E308"
promptText="0" GridPane.columnIndex="1" GridPane.rowIndex="5" />

    </children>
    <rowConstraints>
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
    </rowConstraints>
</GridPane>
<Separator />
<HBox alignment="CENTER" spacing="15.0">
    <children>
        <Button fx:id="closeButton" mnemonicParsing="false"
onAction="#onCloseClick" text="%button.close" />
        <Button fx:id="submitButton" mnemonicParsing="false"
onAction="#onSubmitClick" text="%button.submit" />
    </children>
    <cursor>
        <Cursor fx:constant="DEFAULT" />
    </cursor>
</HBox>
</children>
<padding>
    <Insets bottom="20.0" left="20.0" right="20.0" top="20.0" />
</padding>
</VBox>

```

Листинг 12. give-tool-view.fxml

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.Cursor?>

```

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```

<?import javafx.scene.control.Button?>
<?import javafx.scene.control.ComboBox?>
<?import javafx.scene.control.DatePicker?>
<?import javafx.scene.control.Label?>
<?import javafx.scene.control.Separator?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.ColumnConstraints?>
<?import javafx.scene.layout.GridPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.layout.RowConstraints?>
<?import javafx.scene.layout.VBox?>

<VBox spacing="15.0" xmlns="http://javafx.com/javafx/25"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.GiveToolController">
    <children>
        <GridPane hgap="10.0" vgap="12.0">
            <columnConstraints>
                <ColumnConstraints minWidth="120.0" prefWidth="140.0" />
                <ColumnConstraints hgrow="ALWAYS" minWidth="200.0" />
            </columnConstraints>
            <children>
                <Label text="%label.tool" GridPane.columnIndex="0"
GridPane.rowIndex="0" />
                <ComboBox fx:id="toolComboBox" maxWidth="1.7976931348623157E308"
promptText="%prompt.selectTool" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                <Label text="%label.object" GridPane.columnIndex="0"
GridPane.rowIndex="1" />
                <ComboBox fx:id="objectComboBox" maxWidth="1.7976931348623157E308"
promptText="%prompt.selectObject" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                <Label text="%label.issuedTo" GridPane.columnIndex="0"
GridPane.rowIndex="2" />
                <TextField fx:id="issuedToField" maxWidth="1.7976931348623157E308"
promptText="%prompt.recipientName" GridPane.columnIndex="1" GridPane.rowIndex="2"
/>
                <Label text="%label.issueDate" GridPane.columnIndex="0"
GridPane.rowIndex="3" />
                <DatePicker fx:id="issueDatePicker" GridPane.columnIndex="1"
GridPane.rowIndex="3" />
                <Label text="%label.returnDate" GridPane.columnIndex="0"
GridPane.rowIndex="4" />
                <DatePicker fx:id="returnDatePicker" GridPane.columnIndex="1"
GridPane.rowIndex="4" />
            </children>
            <rowConstraints>

```

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```

        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
    </rowConstraints>
</GridPane>
<Separator />
<HBox alignment="CENTER" spacing="15.0">
    <children>
        <Button fx:id="closeButton" mnemonicParsing="false"
onAction="#onCloseClick" text="%button.close" />
        <Button fx:id="submitButton" mnemonicParsing="false"
onAction="#onSubmitClick" text="%button.issue" />
    </children>
    <cursor>
        <Cursor fx:constant="DEFAULT" />
    </cursor>
</HBox>
</children>
<padding>
    <Insets bottom="20.0" left="20.0" right="20.0" top="20.0" />
</padding>
</VBox>

```

Листинг 13. main-view.fxml

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.Tab?>
<?import javafx.scene.control.TabPane?>
<?import javafx.scene.control.TableView?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.AnchorPane?>
<?import javafx.scene.layout.BorderPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.layout.VBox?>

<BorderPane maxHeight="-Infinity" maxWidth="-Infinity" minHeight="-Infinity"
minWidth="-Infinity" prefHeight="500.0" prefWidth="1000.0"
xmlns="http://javafx.com/javafx/25" xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.MainController">
    <center>

```

```

<TabPane tabClosingPolicy="UNAVAILABLE">
    <tabs>
        <Tab text="%tab.references">
            <content>
                <VBox>
                    <children>
                        <TabPane fx:id="referenceSubTabs"
tabClosingPolicy="UNAVAILABLE">
                            <tabs>
                                <Tab text="%tab.refMaterial" />
                                <Tab text="%tab.refSupplier" />
                                <Tab text="%tab.refTool" />
                                <Tab text="%tab.refProrab" />
                                <Tab text="%tab.refObject" />
                            </tabs>
                        </TabPane>
                        <AnchorPane VBox.vgrow="ALWAYS">
                            <children>
                                <TableView fx:id="referenceTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                            </children>
                        </AnchorPane>
                        <HBox alignment="BOTTOM_LEFT">
                            <children>
                                <Button mnemonicParsing="false"
onAction="#onManageReferencesClick" text="%button.manageReferences">
                                    <HBox.margin>
                                        <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                                    </HBox.margin>
                                </Button>
                                <Button mnemonicParsing="false"
onAction="#onRefreshReferenceClick" text="%button.refresh">
                                    <HBox.margin>
                                        <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                                    </HBox.margin>
                                </Button>
                                <Button mnemonicParsing="false"
onAction="#onSearchReferenceClick" text="%button.search">
                                    <HBox.margin>
                                        <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                                    </HBox.margin>
                                </Button>
                            </children>
                        </HBox>
                    </children>
                </VBox>
            </content>
        </Tab>
    </tabs>
</TabPane>

```



```

        </Button>
        <TextField fx:id="referenceSearchField"
promptText="%filter.prompt">
            <HBox.margin>
                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
            </HBox.margin>
        </TextField>
    </children>
</HBox>
</children>
</VBox>
</content>
</Tab>
<Tab text="%tab.warehouse">
    <content>
        <VBox>
            <children>
                <TabPane fx:id="warehouseSubTabs"
tabClosingPolicy="UNAVAILABLE">
                    <tabs>
                        <Tab text="%tab.whStock" />
                        <Tab text="%tab.whReceipt" />
                        <Tab text="%tab.whInvoice" />
                        <Tab text="%tab.whInvoiceItems" />
                        <Tab text="%tab.whReturn" />
                    </tabs>
                </TabPane>
                <AnchorPane VBox.vgrow="ALWAYS">
                    <children>
                        <TableView fx:id="warehouseTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                    </children>
                </AnchorPane>
                <HBox alignment="BOTTOM_LEFT">
                    <children>
                        <Button mnemonicParsing="false"
onAction="#onRefreshWarehouseClick" text="%button.refresh">
                            <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                            </HBox.margin>
                        </Button>

```

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```

        <Button mnemonicParsing="false"
onAction="#onSearchWarehouseClick" text="%button.search">
            <HBox.margin>
                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
            </HBox.margin>
        </Button>
        <TextField fx:id="warehouseSearchField"
promptText="%filter.prompt">
            <HBox.margin>
                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
            </HBox.margin>
        </TextField>
    </children>
</HBox>
</children>
</VBox>
</content>
</Tab>
<Tab text="%tab.reports">
    <content>
        <VBox>
            <children>
                <TabPane fx:id="reportSubTabs"
tabClosingPolicy="UNAVAILABLE">
                    <tabs>
                        <Tab text="%tab.reportReceipts" />
                        <Tab text="%tab.reportIssues" />
                        <Tab text="%tab.reportDebtors" />
                        <Tab text="%tab.reportToolIssues" />
                    </tabs>
                </TabPane>
                <AnchorPane VBox.vgrow="ALWAYS">
                    <children>
                        <TableView fx:id="reportTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                    </children>
                </AnchorPane>
            <HBox alignment="BOTTOM_LEFT">
                <children>
                    <Button mnemonicParsing="false"
onAction="#onRefreshReportClick" text="%button.refresh">
                        <HBox.margin>

```

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```

                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </Button>
                                <Button mnemonicParsing="false"
onAction="#onSearchReportClick" text="%button.search">
                                <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </Button>
                                <TextField fx:id="reportSearchField"
promptText="%filter.prompt">
                                <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </TextField>
                                </children>
                                </HBox>
                                </children>
                                </VBox>
                                </content>
                                </Tab>
                                </tabs>
                                </TabPane>
                                </center>
                                <right>
                                <VBox prefHeight="200.0" prefWidth="150.0" BorderPane.alignment="CENTER">
                                <children>
                                <Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onPrihodClick" text="%button.receipt">
                                <VBox.margin>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </VBox.margin>
                                <padding>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </padding>
                                </Button>
                                <Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onGiveMaterialClick" text="%button.giveMaterial">
                                <VBox.margin>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </VBox.margin>
                                <padding>

```

```

        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>

<Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onGiveToolClick" text="%button.giveTool">

    <VBox.margin>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </VBox.margin>
    <padding>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>

<Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onReturnToolClick" text="%button.returnTool">

    <VBox.margin>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </VBox.margin>
    <padding>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>
</children>
</VBox>
</right>
</BorderPane>

```

Листинг 14. prihod-view.fxml

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.Tab?>
<?import javafx.scene.control.TabPane?>
<?import javafx.scene.control.TableView?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.AnchorPane?>
<?import javafx.scene.layout.BorderPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.layout.VBox?>

<BorderPane maxHeight="-Infinity" maxWidth="-Infinity" minHeight="-Infinity"
minWidth="-Infinity" prefHeight="500.0" prefWidth="1000.0"
xmlns="http://javafx.com/javafx/25" xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.MainController">

```

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```

<center>
  <TabPane tabClosingPolicy="UNAVAILABLE">
    <tabs>
      <Tab text="%tab.references">
        <content>
          <VBox>
            <children>
              <TabPane fx:id="referenceSubTabs"
tabClosingPolicy="UNAVAILABLE">
                <tabs>
                  <Tab text="%tab.refMaterial" />
                  <Tab text="%tab.refSupplier" />
                  <Tab text="%tab.refTool" />
                  <Tab text="%tab.refProrab" />
                  <Tab text="%tab.refObject" />
                </tabs>
              </TabPane>
              <AnchorPane VBox.vgrow="ALWAYS">
                <children>
                  <TableView fx:id="referenceTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                </children>
              </AnchorPane>
              <HBox alignment="BOTTOM_LEFT">
                <children>
                  <Button mnemonicParsing="false"
onAction="#onManageReferencesClick" text="%button.manageReferences">
                    <HBox.margin>
                      <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                    </HBox.margin>
                  </Button>
                  <Button mnemonicParsing="false"
onAction="#onRefreshReferenceClick" text="%button.refresh">
                    <HBox.margin>
                      <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                    </HBox.margin>
                  </Button>
                  <Button mnemonicParsing="false"
onAction="#onSearchReferenceClick" text="%button.search">
                    <HBox.margin>
                      <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                    </HBox.margin>
                  </Button>
                </children>
              </HBox>
            </children>
          </VBox>
        </content>
      </Tab>
    </tabs>
  </TabPane>
</center>

```

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```

        </HBox.margin>
        </Button>
        <TextField fx:id="referenceSearchField"
promptText="%filter.prompt">
            <HBox.margin>
                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
            </HBox.margin>
        </TextField>
    </children>
</HBox>
</children>
</VBox>
</content>
</Tab>
<Tab text="%tab.warehouse">
    <content>
        <VBox>
            <children>
                <TabPane fx:id="warehouseSubTabs"
tabClosingPolicy="UNAVAILABLE">
                    <tabs>
                        <Tab text="%tab.whStock" />
                        <Tab text="%tab.whReceipt" />
                        <Tab text="%tab.whInvoice" />
                        <Tab text="%tab.whInvoiceItems" />
                        <Tab text="%tab.whReturn" />
                    </tabs>
                </TabPane>
                <AnchorPane VBox.vgrow="ALWAYS">
                    <children>
                        <TableView fx:id="warehouseTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                    </children>
                </AnchorPane>
                <HBox alignment="BOTTOM_LEFT">
                    <children>
                        <Button mnemonicParsing="false"
onAction="#onRefreshWarehouseClick" text="%button.refresh">
                            <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
                            </HBox.margin>
                        </Button>

```

```

<Button mnemonicParsing="false"
onAction="#onSearchWarehouseClick" text="%button.search">
    <HBox.margin>
        <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
    </HBox.margin>
</Button>
<TextField fx:id="warehouseSearchField"
promptText="%filter.prompt">
    <HBox.margin>
        <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />
    </HBox.margin>
</TextField>
</children>
</HBox>
</children>
</VBox>
</content>
</Tab>
<Tab text="%tab.reports">
    <content>
        <VBox>
            <children>
                <TabPane fx:id="reportSubTabs"
tabClosingPolicy="UNAVAILABLE">
                    <tabs>
                        <Tab text="%tab.reportReceipts" />
                        <Tab text="%tab.reportIssues" />
                        <Tab text="%tab.reportDebtors" />
                        <Tab text="%tab.reportToolIssues" />
                    </tabs>
                </TabPane>
                <AnchorPane VBox.vgrow="ALWAYS">
                    <children>
                        <TableView fx:id="reportTableView"
AnchorPane.bottomAnchor="0.0" AnchorPane.leftAnchor="0.0"
AnchorPane.rightAnchor="0.0" AnchorPane.topAnchor="0.0" />
                    </children>
                </AnchorPane>
            <HBox alignment="BOTTOM_LEFT">
                <children>
                    <Button mnemonicParsing="false"
onAction="#onRefreshReportClick" text="%button.refresh">
                        <HBox.margin>

```

```

                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </Button>
                                <Button mnemonicParsing="false"
onAction="#onSearchReportClick" text="%button.search">
                                <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </Button>
                                <TextField fx:id="reportSearchField"
promptText="%filter.prompt">
                                <HBox.margin>
                                <Insets bottom="10.0" left="10.0" right="10.0"
top="10.0" />

                                </HBox.margin>
                                </TextField>
                                </children>
                                </HBox>
                                </children>
                                </VBox>
                                </content>
                                </Tab>
                                </tabs>
                                </TabPage>
                                </center>
                                <right>
                                <VBox prefHeight="200.0" prefWidth="150.0" BorderPane.alignment="CENTER">
                                <children>
                                <Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onPrihodClick" text="%button.receipt">
                                <VBox.margin>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </VBox.margin>
                                <padding>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </padding>
                                </Button>
                                <Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onGiveMaterialClick" text="%button.giveMaterial">
                                <VBox.margin>
                                <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
                                </VBox.margin>
                                <padding>

```



```

        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>

<Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onGiveToolClick" text="%button.giveTool">

    <VBox.margin>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </VBox.margin>
    <padding>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>

<Button maxWidth="1.7976931348623157E308" mnemonicParsing="false"
onAction="#onReturnToolClick" text="%button.returnTool">

    <VBox.margin>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </VBox.margin>
    <padding>
        <Insets bottom="5.0" left="5.0" right="5.0" top="5.0" />
    </padding>
</Button>
</children>
</VBox>
</right>
</BorderPane>

```

Листинг 15. reference-editor-view.fxml

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.ComboBox?>
<?import javafx.scene.control.Label?>
<?import javafx.scene.control.Separator?>
<?import javafx.scene.control.Tab?>
<?import javafx.scene.control.TabPane?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.ColumnConstraints?>
<?import javafx.scene.layout.GridPane?>
<?import javafx.scene.layout.HBox?>
<?import javafx.scene.layout.RowConstraints?>
<?import javafx.scene.layout.VBox?>

```

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```

<VBox spacing="15.0" xmlns="http://javafx.com/javafx/25"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.ReferenceEditorController">
    <children>
        <TabPane fx:id="referenceTabPane" tabClosingPolicy="UNAVAILABLE">
            <tabs>
                <Tab text="%tab.refMaterial">
                    <content>
                        <VBox spacing="12.0">
                            <children>
                                <GridPane hgap="10.0" vgap="10.0">
                                    <columnConstraints>
                                        <ColumnConstraints minWidth="120.0" prefWidth="140.0"
/>
                                        <ColumnConstraints hgrow="ALWAYS" minWidth="220.0" />
                                    </columnConstraints>
                                    <children>
                                        <Label text="%label.name" GridPane.columnIndex="0"
GridPane.rowIndex="0" />
                                        <TextField fx:id="materialNameField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                                        <Label text="%label.article" GridPane.columnIndex="0"
GridPane.rowIndex="1" />
                                        <TextField fx:id="materialArticleField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                                        <Label text="%label.unit" GridPane.columnIndex="0"
GridPane.rowIndex="2" />
                                        <TextField fx:id="materialUnitField"
maxWidth="1.7976931348623157E308" promptText="%prompt.unitExample"
GridPane.columnIndex="1" GridPane.rowIndex="2" />
                                        <Label text="%label.writeOffRate"
GridPane.columnIndex="0" GridPane.rowIndex="3" />
                                        <TextField fx:id="materialWriteOffRateField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="3" />
                                    </children>
                                    <rowConstraints>
                                        <RowConstraints />
                                        <RowConstraints />
                                        <RowConstraints />
                                        <RowConstraints />
                                    </rowConstraints>
                                </GridPane>
                                <HBox alignment="CENTER_RIGHT">
                                    <children>

```

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```

        <Button fx:id="materialSubmitButton"
mnemonicParsing="false" onAction="#onMaterialSubmitClick" text="%button.add" />
        </children>
    </HBox>
</children>
<padding>
    <Insets bottom="10.0" left="10.0" right="10.0" top="10.0"
/>

    </padding>
</VBox>
</content>
</Tab>
<Tab text="%tab.refSupplier">
    <content>
        <VBox spacing="12.0">
            <children>
                <GridPane hgap="10.0" vgap="10.0">
                    <columnConstraints>
                        <ColumnConstraints minWidth="120.0" prefWidth="140.0"
/>

                        <ColumnConstraints hgrow="ALWAYS" minWidth="220.0" />
                    </columnConstraints>
                    <children>
                        <Label text="%label.name" GridPane.columnIndex="0"
GridPane.rowIndex="0" />

                        <TextField fx:id="supplierNameField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                        <Label text="%label.contactPerson"
GridPane.columnIndex="0" GridPane.rowIndex="1" />
                        <TextField fx:id="supplierContactField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                        <Label text="%label.phone" GridPane.columnIndex="0"
GridPane.rowIndex="2" />
                        <TextField fx:id="supplierPhoneField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="2" />
                        <Label text="%label.email" GridPane.columnIndex="0"
GridPane.rowIndex="3" />
                        <TextField fx:id="supplierEmailField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="3" />
                        <Label text="%label.address" GridPane.columnIndex="0"
GridPane.rowIndex="4" />
                        <TextField fx:id="supplierAddressField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="4" />
                    </children>
                    <rowConstraints>

```

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```

        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
        <RowConstraints />
    </rowConstraints>
</GridPane>
<HBox alignment="CENTER_RIGHT">
    <children>
        <Button fx:id="supplierSubmitButton"
mnemonicParsing="false" onAction="#onSupplierSubmitClick" text="%button.add" />
    </children>
</HBox>
</children>
<padding>
    <Insets bottom="10.0" left="10.0" right="10.0" top="10.0"
/>

    </padding>
</VBox>
</content>
</Tab>
<Tab text="%tab.refProrab">
    <content>
        <VBox spacing="12.0">
            <children>
                <GridPane hgap="10.0" vgap="10.0">
                    <columnConstraints>
                        <ColumnConstraints minWidth="120.0" prefWidth="140.0"
/>

                        <ColumnConstraints hgrow="ALWAYS" minWidth="220.0" />
                    </columnConstraints>
                    <children>
                        <Label text="%label.name" GridPane.columnIndex="0"
GridPane.rowIndex="0" />
                        <TextField fx:id="prorabNameField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                        <Label text="%label.phone" GridPane.columnIndex="0"
GridPane.rowIndex="1" />
                        <TextField fx:id="prorabPhoneField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                    </children>
                    <rowConstraints>
                        <RowConstraints />
                        <RowConstraints />
                    </rowConstraints>

```

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```

        </GridPane>
        <HBox alignment="CENTER_RIGHT">
            <children>
                <Button fx:id="prorabSubmitButton"
mnemonicParsing="false" onAction="#onProrabSubmitClick" text="%button.add" />
            </children>
        </HBox>
    </children>
    <padding>
        <Insets bottom="10.0" left="10.0" right="10.0" top="10.0"
/>

    </padding>
</VBox>
</content>
</Tab>
<Tab fx:id="objectTab" text="%tab.refObject">
    <content>
        <VBox spacing="12.0">
            <children>
                <GridPane hgap="10.0" vgap="10.0">
                    <columnConstraints>
                        <ColumnConstraints minWidth="120.0" prefWidth="140.0"
/>

                        <ColumnConstraints hgrow="ALWAYS" minWidth="220.0" />
                    </columnConstraints>
                    <children>
                        <Label text="%label.name" GridPane.columnIndex="0"
GridPane.rowIndex="0" />

                        <TextField fx:id="objectNameField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                        <Label text="%label.address" GridPane.columnIndex="0"
GridPane.rowIndex="1" />

                        <TextField fx:id="objectAddressField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                        <Label text="%label.prorab" GridPane.columnIndex="0"
GridPane.rowIndex="2" />

                        <ComboBox fx:id="objectProrabComboBox"
maxWidth="1.7976931348623157E308" promptText="%prompt.selectProrab"
GridPane.columnIndex="1" GridPane.rowIndex="2" />
                    </children>
                    <rowConstraints>
                        <RowConstraints />
                        <RowConstraints />
                        <RowConstraints />
                    </rowConstraints>

```

```

        </GridPane>
        <HBox alignment="CENTER_RIGHT">
            <children>
                <Button fx:id="objectSubmitButton"
mnemonicParsing="false" onAction="#onObjectSubmitClick" text="%button.add" />
            </children>
        </HBox>
    </children>
    <padding>
        <Insets bottom="10.0" left="10.0" right="10.0" top="10.0"
/>

    </padding>
</VBox>
</content>
</Tab>
<Tab text="%tab.refTool">
    <content>
        <VBox spacing="12.0">
            <children>
                <GridPane hgap="10.0" vgap="10.0">
                    <columnConstraints>
                        <ColumnConstraints minWidth="120.0" prefWidth="140.0"
/>

                        <ColumnConstraints hgrow="ALWAYS" minWidth="220.0" />
                    </columnConstraints>
                    <children>
                        <Label text="%label.name" GridPane.columnIndex="0"
GridPane.rowIndex="0" />

                        <TextField fx:id="toolNameField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                        <Label text="%label.inventoryNumber"
GridPane.columnIndex="0" GridPane.rowIndex="1" />
                        <TextField fx:id="toolInventoryField"
maxWidth="1.7976931348623157E308" GridPane.columnIndex="1" GridPane.rowIndex="1" />
                        <Label text="%label.condition"
GridPane.columnIndex="0" GridPane.rowIndex="2" />
                        <TextField fx:id="toolConditionField"
maxWidth="1.7976931348623157E308" promptText="%prompt.condition"
GridPane.columnIndex="1" GridPane.rowIndex="2" />
                    </children>
                    <rowConstraints>
                        <RowConstraints />
                        <RowConstraints />
                        <RowConstraints />
                    </rowConstraints>

```

```

        </GridPane>
        <HBox alignment="CENTER_RIGHT">
            <children>
                <Button fx:id="toolSubmitButton"
mnemonicParsing="false" onAction="#onToolSubmitClick" text="%button.add" />
            </children>
        </HBox>
    </children>
    <padding>
        <Insets bottom="10.0" left="10.0" right="10.0" top="10.0"
/>
    </padding>
</VBox>
</content>
</Tab>
</tabs>
</TabPage>
<Separator />
<HBox alignment="CENTER" spacing="15.0">
    <children>
        <Button mnemonicParsing="false" onAction="#onCloseClick"
text="%button.close" />
    </children>
</HBox>
</children>
<padding>
    <Insets bottom="20.0" left="20.0" right="20.0" top="20.0" />
</padding>
</VBox>

```

Листинг 16. return-tool-view.fxml

```

<?xml version="1.0" encoding="UTF-8"?>

<?import javafx.geometry.Insets?>
<?import javafx.scene.Cursor?>
<?import javafx.scene.control.Button?>
<?import javafx.scene.control.ComboBox?>
<?import javafx.scene.control.DatePicker?>
<?import javafx.scene.control.Label?>
<?import javafx.scene.control.Separator?>
<?import javafx.scene.control.TextField?>
<?import javafx.scene.layout.ColumnConstraints?>
<?import javafx.scene.layout.GridPane?>
<?import javafx.scene.layout.HBox?>

```

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```

<?import javafx.scene.layout.RowConstraints?>
<?import javafx.scene.layout.VBox?>

<VBox spacing="15.0" xmlns="http://javafx.com/javafx/25"
xmlns:fx="http://javafx.com/fxml/1"
fx:controller="ru.kochedykovdev.laba7.controllers.ReturnToolController">
    <children>
        <GridPane hgap="10.0" vgap="12.0">
            <columnConstraints>
                <ColumnConstraints minWidth="120.0" prefWidth="140.0" />
                <ColumnConstraints hgrow="ALWAYS" minWidth="200.0" />
            </columnConstraints>
            <children>
                <Label text="%label.issue" GridPane.columnIndex="0"
GridPane.rowIndex="0" />
                <ComboBox fx:id="toolIssueComboBox" maxWidth="1.7976931348623157E308"
promptText="%prompt.issueSelect" GridPane.columnIndex="1" GridPane.rowIndex="0" />
                <Label text="%label.actualReturnDate" GridPane.columnIndex="0"
GridPane.rowIndex="1" />
                <DatePicker fx:id="actualReturnDatePicker" GridPane.columnIndex="1"
GridPane.rowIndex="1" />
                <Label text="%label.condition" GridPane.columnIndex="0"
GridPane.rowIndex="2" />
                <TextField fx:id="conditionField" maxWidth="1.7976931348623157E308"
promptText="%prompt.condition" GridPane.columnIndex="1" GridPane.rowIndex="2" />
            </children>
            <rowConstraints>
                <RowConstraints />
                <RowConstraints />
                <RowConstraints />
            </rowConstraints>
        </GridPane>
        <Separator />
        <HBox alignment="CENTER" spacing="15.0">
            <children>
                <Button fx:id="closeButton" mnemonicParsing="false"
onAction="#onCloseClick" text="%button.close" />
                <Button fx:id="submitButton" mnemonicParsing="false"
onAction="#onSubmitClick" text="%button.acceptReturn" />
            </children>
            <cursor>
                <Cursor fx:constant="DEFAULT" />
            </cursor>
        </HBox>
    </children>

```

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```

    <padding>
      <Insets bottom="20.0" left="20.0" right="20.0" top="20.0" />
    </padding>
  </VBox>

```

Листинг 16. logback.xml

```

<?xml version="1.0" encoding="UTF-8"?>
<configuration>
  <appender name="CONSOLE" class="ch.qos.logback.core.ConsoleAppender">
    <encoder>
      <pattern>%d{HH:mm:ss.SSS} [%level] %logger - %msg%n</pattern>
    </encoder>
  </appender>

  <appender name="FILE" class="ch.qos.logback.core.FileAppender">
    <file>logs/app.log</file>
    <append>true</append>
    <encoder>
      <pattern>%d{yyyy-MM-dd HH:mm:ss} [%level] %logger - %msg%n</pattern>
    </encoder>
  </appender>

  <root level="INFO">
    <appender-ref ref="CONSOLE"/>
    <appender-ref ref="FILE"/>
  </root>
</configuration>

```

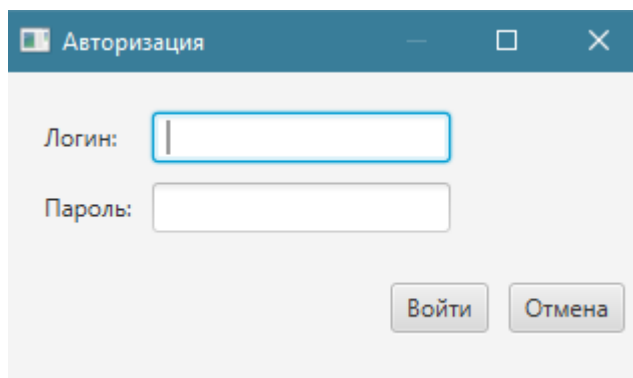


Рисунок 1 – Модальное окно авторизации

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Приход материала

Материал: Выберите материал

Поставщик: Выберите поставщика

Количество: 0

Цена за единицу: 0

Дата прихода: 04.06.2026

Закрыть Провести

Рисунок 4 – Приход материала

Выдача материала

№ накладной: НК-001

Объект: Институт МИВлГУ (капремонт)

Дата выдачи: 04.06.2026

Материал: Цемент М500

Количество: 100

Цена на момент выдачи: 50

Закрыть Провести

Рисунок 5 – Выдача материала и доступная кнопка для проведения

```
00:35:35.925 [INFO] ru.kochedykovdev.laba7.HelloApplication - Приложение запущено
00:35:35.930 [INFO] ru.kochedykovdev.laba7.HelloApplication - Локаль: ru_RU, ресурсы: ru
WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
WARNING: sun.misc.Unsafe::allocateMemory has been called by com.sun.marlin.OffHeapArray (file:/C:/Users/Tee/.m2/repository/org/openjfx/javafx-graphics/21.0.6/javafx-graphics-21.0.6-win.jar)
WARNING: Please consider reporting this to the maintainers of class com.sun.marlin.OffHeapArray
WARNING: sun.misc.Unsafe::allocateMemory will be removed in a future release
00:36:09.014 [INFO] ru.kochedykovdev.laba7.util.DBHelper - Подключение к jdbc:postgresql://localhost:5432/postgres пользователем egor
00:36:09.229 [INFO] ru.kochedykovdev.laba7.util.DBHelper - Соединение установлено, схема: kursovaya4sem
00:36:09.230 [INFO] ru.kochedykovdev.laba7.HelloApplication - Успешное подключение для egor
инн. 04, 2026 12:36:09 AM javafx.fxml.FXMLLoader$ValueElement processValue
WARNING: Loading FXML document with JavaFX API of version 25 by JavaFX runtime of version 21.0.6
00:36:09.411 [INFO] ru.kochedykovdev.laba7.util.DBHelper - Соединение установлено, схема: kursovaya4sem
```

Рисунок 6 – Работа логгера в терминале

772	2026-06-04 00:15:10	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	✓ 196 ^
773	2026-06-04 00:15:11	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
774	2026-06-04 00:15:11	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
775	2026-06-04 00:15:12	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение закрыто	
776	2026-06-04 00:15:12	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Приложение завершено	
777	2026-06-04 00:18:57	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Приложение запущено	
778	2026-06-04 00:18:57	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Локаль: ru_RU, ресурсы: ru	
779	2026-06-04 00:18:59	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Пользователь отменил вход, выход	
780	2026-06-04 00:18:59	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Приложение завершено	
781	2026-06-04 00:35:35	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Приложение запущено	
782	2026-06-04 00:35:35	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Локаль: ru_RU, ресурсы: ru	
783	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Подключение к jdbc:postgresql://localhost:5432/postgres	п
784	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
785	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Успешное подключение для egor	
786	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
787	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
788	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
789	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
790	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
791	2026-06-04 00:36:09	[INFO]	ru.kochedykovdev.laba7.HelloApplication	- Главное окно открыто	
792	2026-06-04 00:36:34	[INFO]	ru.kochedykovdev.laba7.controllers.MainController	- Открытие окна: dialog.references	
793	2026-06-04 00:36:34	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
794	2026-06-04 00:36:42	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
795	2026-06-04 00:36:42	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	
796	2026-06-04 00:36:42	[INFO]	ru.kochedykovdev.laba7.util.DBHelper	- Соединение установлено, схема: kursovaya4sem	

Рисунок 7 – Работа логгера в файле app.log

Вывод: В ходе лабораторной работы я изучил способы интернационализации приложения